Continuous Probability Distributions

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Objectives

- Understand the properties of continuous random variables, and why the probability of any single value is zero.
- Read a probability density correctly: the **area** under it is a probability, its **height** is not.
- Identify the characteristics of the normal probability distribution.
- Calculate normal probabilities for every shape of interval, and interpret z-scores, knowing when the standard normal applies to them and when it does not.
- Work backwards from a probability to a value: percentiles, targets, reorder points.
- Revisit the empirical rule, and judge whether real data are normal enough for the normal model to be trusted.
- Use the normal distribution to approximate the binomial distribution, including the continuity correction for every way the question can be worded.
- Identify the characteristics of the exponential probability distribution, keep its rate and its mean straight, and calculate exponential probabilities.
- Identify the characteristics of the uniform probability distribution and calculate uniform probabilities.

Chapter 5 put a probability on every value a **count** could take. This chapter does the same for a **measurement**: a waiting time, a flight time, a month's sales, a quarter's growth rate. The change sounds small. It is not. A measurement has infinitely many possible values in any interval, so we can no longer list them and attach a probability to each. Probability becomes an **area under a curve**, and every question in this chapter turns into "which area, and how do I get it?"

The normal distribution, the bell curve from Chapter 3, is the centre of the chapter and of the rest of the course. Every confidence interval and hypothesis test in Chapters 8 through 15 is built on it. So the habits worth building now are the ones that carry: name the shape of the area you want, sketch it, compute it as a difference of two cumulative probabilities, and check the answer against the picture.

As in Chapter 5, you will not grind through these by hand on an exam. You will be handed output and asked which number answers the question and what it means. The tables in this note are the skill.

Continuous Random Variables

Continuous random variables are outcomes that take on any numerical value in an interval, as determined by conducting an experiment. They are usually *measured* rather than counted: time, distance, weight, money treated as a continuum, rates of growth.

Consider the flight time of American Airlines flight 2348 from *Las Vegas* to *Washington, DC*. One evening it completed the trip in 4 hours, 24 minutes, and 33.3227 seconds, or 4.4092563 hours. A more precise clock would add more digits. There is no last digit.

What is the probability that the flight takes *exactly* 4.4 hours? Between any two flight times, however close, there are infinitely many others, and 4.4 hours is one point among them. Its probability is zero:

$$P(x = 4.4) = 0.$$

That is not the same as saying a 4.4-hour flight is impossible. It says that to get a probability bigger than zero, we have to ask about an **interval**: between 4.3 and 4.5 hours, or less than 4.4 hours.

A practical consequence: for a continuous random variable, whether the end point is included makes no difference.

$$P(x \leq 4.4) = P(x < 4.4) + P(x = 4.4) = P(x < 4.4).$$

Contrast this with Chapter 5, where "fewer than 4" and "4 or fewer" were different events with different probabilities and confusing them was the chapter's most common mistake. With a measurement, "less than" and "at most" are the same question. That mistake goes away here, and comes back the moment we use a continuous curve to approximate a count, later in this note.

Density is not probability

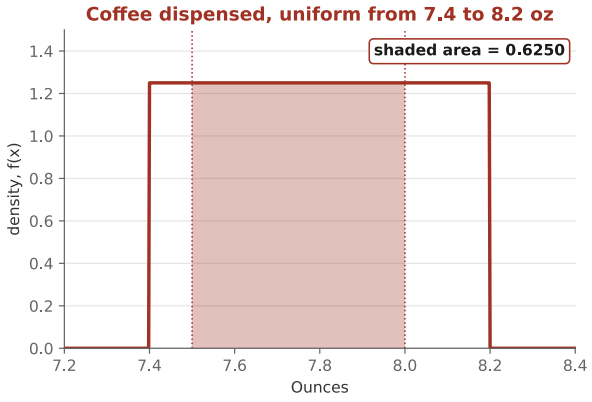
If a single value has probability zero, what does a curve for a continuous random variable show?

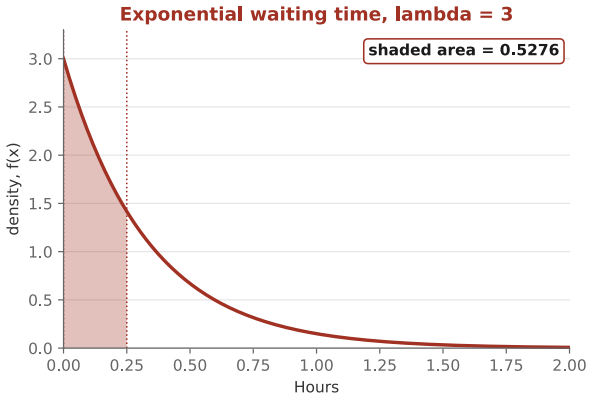
A **probability density function**, $f(x)$, is a curve with two properties:

1. It is never negative: $f(x) \geq 0$.
2. The total **area** under it is 1.

The probability that x falls in an interval is the **area under the curve over that interval**. The **height** of the curve, $f(x)$, is a *density*: it says where values are more or less concentrated, but it is not the probability of anything.

The difference matters because densities can be larger than 1, and probabilities cannot. Two examples from later in the chapter:



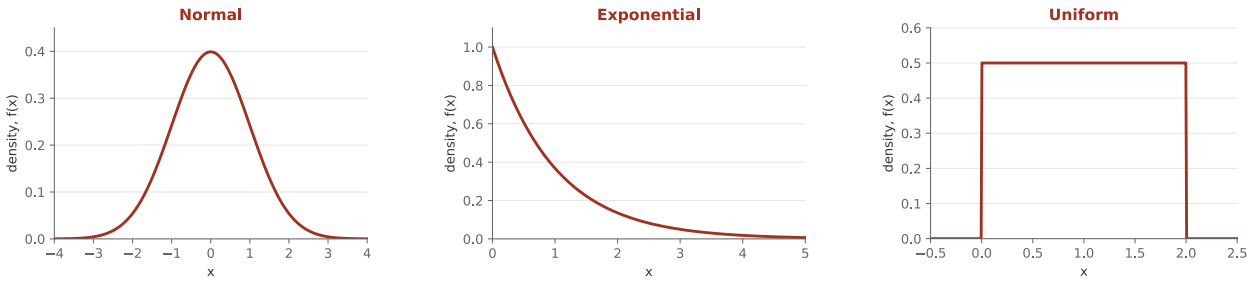


The coffee machine's density is 1.25 everywhere between 7.4 and 8.2 ounces, because the rectangle is only 0.8 wide and its area has to be 1. The exponential curve starts at a height of 3. Both are perfectly good distributions. The shaded areas, 0.6250 and 0.5276, are the probabilities.

A chart of a continuous distribution whose vertical axis says "Probability" is mislabelled, and the mislabel teaches the wrong idea: that you can read a probability off the curve by looking at its height. Every density chart in these notes labels the axis **density**, and every probability is shown as a shaded area. When you ask Claude for one of these charts, check the axis.

Three shapes

This chapter covers three continuous distributions. They are not the only ones, but between them they cover most of what a manager will meet.



- The **normal** distribution describes measurements that cluster around a centre, with large departures in either direction rare: fill weights on a production line, test scores, forecast errors.
- The **exponential** distribution describes waiting times between events, where short waits are common and long ones rarer: time between customer arrivals, time between machine failures.
- The **uniform** distribution describes a measurement that is equally likely to fall anywhere in a range: the wait for a shuttle that runs on a fixed schedule when you arrive at a random moment.

The Normal Distribution

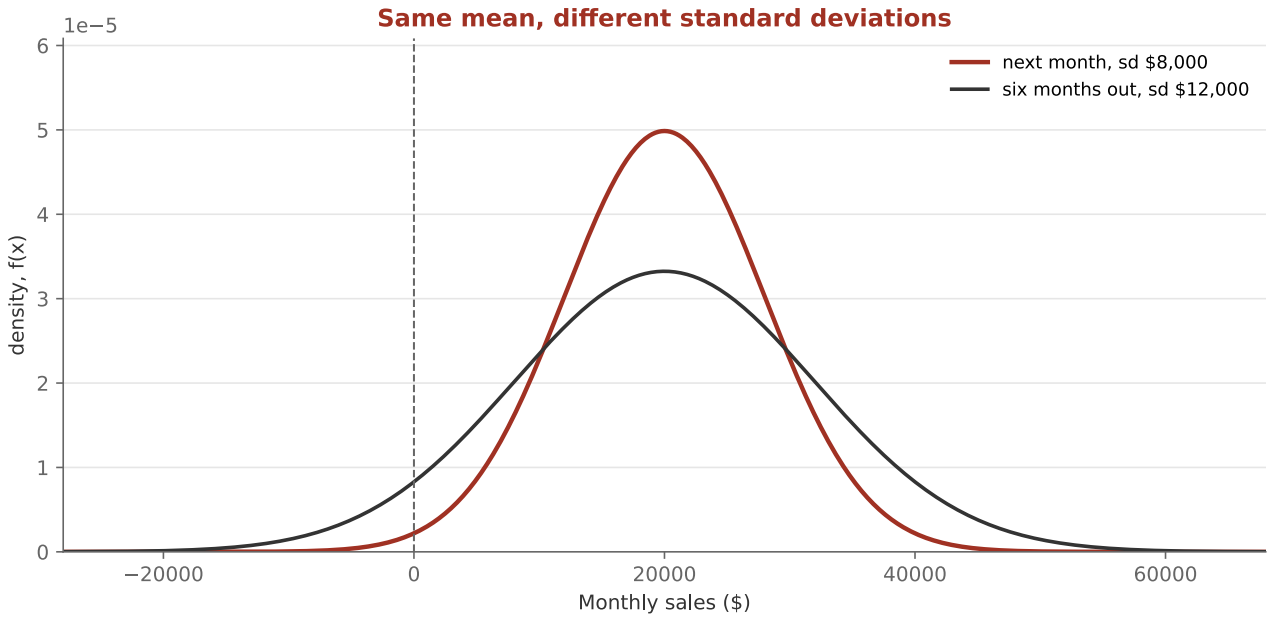
Characteristics of the normal probability distribution

The **normal probability distribution** is a continuous distribution that is bell-shaped and symmetric around its mean. Its characteristics:

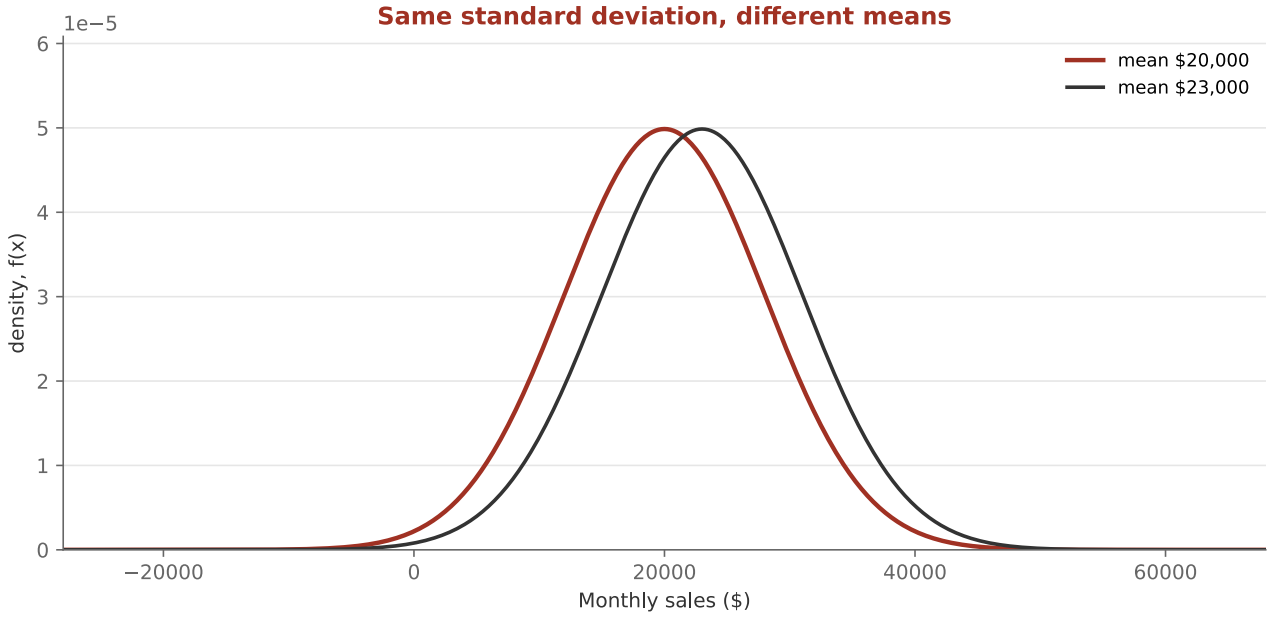
- The distribution is bell-shaped and symmetric around its mean, μ.
- Because it is symmetric, the mean and the median are the same value.
- Values near the mean, where the curve is tallest, are more likely than values toward the ends, where the curve is lower.
- The total area under the curve is 1.
- Because the curve is symmetric around the mean, the area to the left of the mean is 0.5, and so is the area to the right.
- The left and right ends extend indefinitely, never quite touching the horizontal axis.

What the two parameters do

Suppose we model next month's sales at a retail shop as normal with a mean of $20,000 and a standard deviation of $8,000. Because of uncertainty in the economy, sales six months out are modelled as normal with the same mean and a standard deviation of $12,000. What does the larger standard deviation do?



A larger standard deviation spreads the curve out and lowers it. The area still has to be 1, so a wider curve must be a shorter one. Now suppose, five months later, the economy has improved: the standard deviation for next month is back to $8,000, and the mean is $23,000.



Changing the mean slides the curve along the axis without changing its shape.

Insight: for a normal distribution, the two parameters μ and σ completely determine the curve. Know them and you know every probability.

Look at the dashed line at zero in the first chart. The curve does not stop there. With a standard deviation of $12,000, this model puts a 4.8% probability on **negative** sales, and even at $8,000 it is 0.62%. A shop cannot sell less than nothing.

The normal distribution is a model, and its tails run forever in both directions. That is harmless when the impossible region is many standard deviations away, and misleading when it is not. You met the same problem in Chapter 3, where three standard deviations below mean Nevada income was a negative number. Before using a normal model, ask whether the variable can take every value the curve allows.

The **normal probability density function** is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2},$$

where $e \approx 2.71828$, $\pi \approx 3.14159$, μ is the mean of the distribution, σ is its standard deviation, and x is any real number.

You will never need to evaluate this formula. It is here so that you know the curve is a specific function of μ and σ and nothing else, and because it is what software is evaluating when you ask for a normal probability.

The z-score and the standard normal distribution

The ***z*-score** is the number of standard deviations a value x lies from the mean of its distribution:

$$z = \frac{x - \mu}{\sigma}.$$

z is negative for values below the mean, positive for values above it, and zero at the mean. It has no units: minutes, dollars, and inches all become "standard deviations."

The **standard normal distribution** is the normal distribution with $\mu = 0$ and $\sigma = 1$.

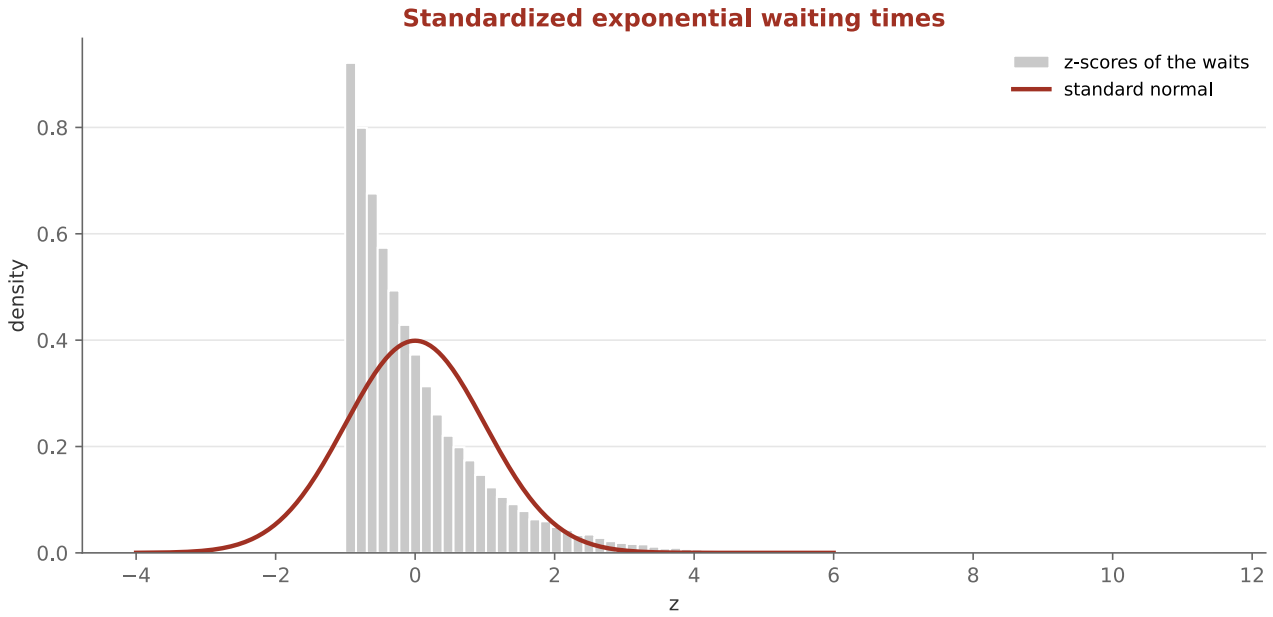
If x is normally distributed, then its z-scores follow the standard normal distribution. Standardizing is just a change of units, like converting feet to inches: the curve keeps its shape, and the axis is relabelled. That is why one table, or one Excel function, can serve every normal distribution there is.

Read the condition in that sentence again: **if x is normally distributed.** Standardizing *any* data gives z-scores with a mean of 0 and a standard deviation of 1. It does **not** make them normal. The z-scores of skewed data are exactly as skewed as the data.

Here is the proof by example. Take 100,000 waiting times from an exponential distribution, the right-skewed shape from the section above, and standardize them.

```
mean of z = -0.0000,  sd of z = 1.0000
```

event	share observed	standard normal says
z < −2	0.0000	0.0228
z < −1	0.0028	0.1587
z > 1	0.1359	0.1587
z > 2	0.0503	0.0228



The mean is 0 and the standard deviation is 1, as promised. Everything else is wrong. The standard normal puts 15.9% of values below $z = -1$; these waits put almost none there, 0.29%, and in theory none at all: a wait cannot be shorter than zero, and zero is exactly one standard deviation below the mean. Above $z = 2$ the standard normal puts 2.3%, and the waits put 5.0%, more than twice as much. Every z-score was computed correctly. A normal probability attached to any of them would be wrong.

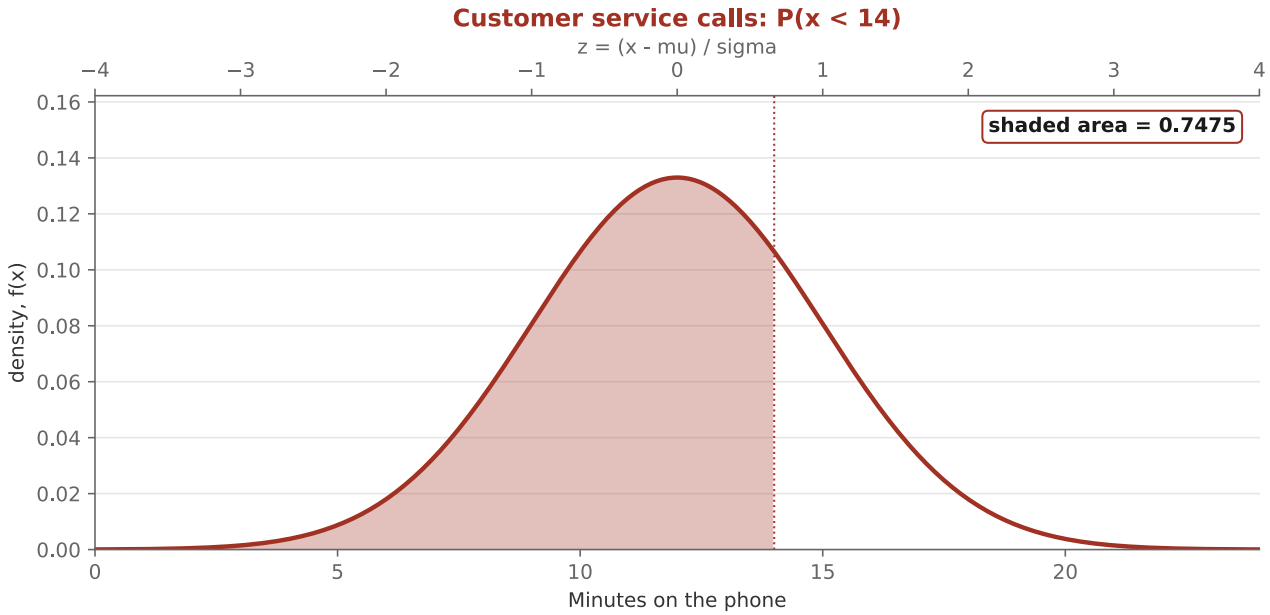
Chapter 3 ran into the same thing with inflation: the z-scores were fine, and the bell-curve percentages did not fit them, because inflation is skewed. A z-score tells you *how far*. Only the shape of the distribution tells you *how likely*.

Calculating normal probabilities

The time customers spend on the phone with a cable company's customer service line follows a normal distribution with a mean of 12 minutes and a standard deviation of 3 minutes. What is the probability that the next caller spends **14 minutes or less** on the phone?

	next caller
event	X < 14
shape	left tail
z lower	-inf
z upper	0.666667
P(X <= upper)	0.747507
P(X <= lower)	0.0
computed as	P(X <= 14)
probability	0.747507
Excel	=NORM.DIST(14, 12, 3, TRUE)

$$z = \frac{x - \mu}{\sigma} = \frac{14 - 12}{3} = 0.6667 \qquad P(x \leq 14) = P(z \leq 0.6667) = 0.7475$$



There is a 74.75% chance the next caller is on the phone 14 minutes or less. The chart carries two scales: the original minutes along the bottom and the z-scores along the top. The area is the same area whichever scale you read it on. That is all "converting to the standard normal" means.

Every normal probability is a difference of two cumulative probabilities. Software gives you one thing, $P(X \le x)$, the area to the *left* of x. Everything else follows from the total area being 1:

You want	Compute	In Excel
$P(X < b)$	$P(X \le b)$	`=NORM.DIST(b, mu, sigma, TRUE)`
$P(X > a)$	$1 - P(X \le a)$	`=1-NORM.DIST(a, mu, sigma, TRUE)`
$P(a < X < b)$	$P(X \le b) - P(X \le a)$	`=NORM.DIST(b,...)-NORM.DIST(a,...)`

`NORM.S.DIST(z, TRUE)` does the same for the standard normal, given a z-score instead of x, μ, and σ. With `FALSE` as the last argument, both functions return the *height* of the curve, which, as above, is not a probability. You will almost never want `FALSE` in this chapter.

The four basic questions about the call centre, in one table:

question	event	shape	z lower	z upper	computed as	probability
1	X < 14	left tail	-inf	0.666667	P(X <= 14)	0.7475
2	X > 14	right tail	0.666667	+inf	1 - P(X <= 14)	0.2525
3	X < 8.5	left tail	-inf	−1.166667	P(X <= 8.5)	0.1217
4	X > 8.5	right tail	−1.166667	+inf	1 - P(X <= 8.5)	0.8783

Rows 1 and 2 sum to 1, and so do rows 3 and 4: each pair is an event and its complement. Run that check every time.

Symmetry is a shortcut, not a detour

Return to the shop with sales normal, mean \$20,000 and standard deviation \$8,000. What is the probability that sales exceed \$16,000?

$$z = \frac{16,000 - 20,000}{8,000} = -0.5 \qquad P(z > -0.5) = P(z < 0.5) = 0.6915$$

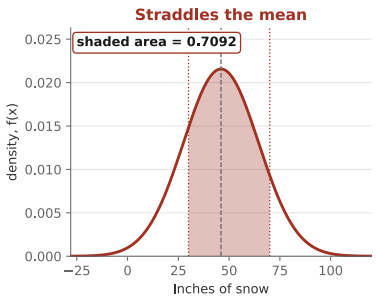
The area to the right of -0.5 is the mirror image of the area to the left of $+0.5$. One line. Sales exceed \$16,000 with probability 69.15%.

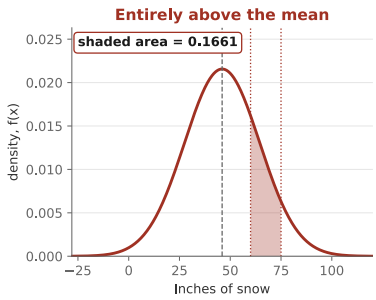
Your textbook prints cumulative normal tables in Appendix A, and some questions are written with them in mind. We will not use them, for the reason we did not use the binomial tables in Chapter 5: they hold the same numbers as `NORM.DIST` for a limited menu of z-scores, rounded to two decimals.

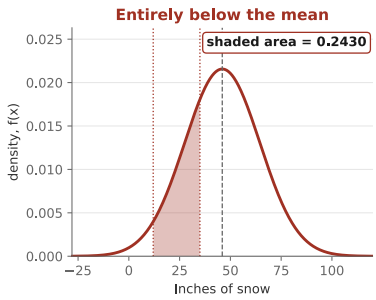
Five shapes of interval

Your textbook's author compiled the annual snowfall for Minneapolis, Minnesota, from 1884 to 2017. It follows a normal distribution reasonably well, with a mean of 46 inches and a standard deviation of 18.5 inches. If history is a good guide, what is the probability that next winter brings each of the following?

question	event	shape	z lower	z upper	P(X <= upper)	P(X <= lower)	probability
1	X < 20	left tail	-inf	−1.405405	0.0800	0.0000	0.0800
2	X > 70	right tail	1.297297	+inf	1.0000	0.9027	0.0973
3	30 < X < 70	straddles the mean	−0.864865	1.297297	0.9027	0.1936	0.7092
4	60 < X < 75	entirely above the mean	0.756757	1.567568	0.9415	0.7754	0.1661
5	12 < X < 35	entirely below the mean	−1.837838	−0.594595	0.2761	0.0330	0.2430







Name the shape before you compute. There are only five: a left tail, a right tail, an interval straddling the mean, an interval entirely above the mean, and an interval entirely below it. Every one of them is computed the same way, $P(X \leq \text{upper}) - P(X \leq \text{lower})$, with the missing end of a tail treated as 0 or 1. The only thing that changes is the picture.

So draw the picture. Your textbook's author reports that students accuse him of trickery when the interval falls entirely on one side of the mean, as in the 60-to-75-inch question. There is no trick. The common error is to treat the area from the mean to 75 inches as the answer, or to add 0.5 to something. A sketch with the mean marked makes both errors visible before they happen.

Reading the output

A Strip resort's revenue manager models the weekend nightly rate for a standard room as normal with a mean of \$210 and a standard deviation of \$35. She asks Claude for cumulative probabilities and receives the table below. Using only the table, find the probability that a randomly chosen weekend night is priced:

(a) below \$175

(b) above \$245

(c) between \$175 and \$245

(d) between \$220 and \$280

(e) above \$140

(f) exactly \$210

nightly rate x (\$)	z	P(X <= x)
140	−2.0000	0.0228
175	−1.0000	0.1587
210	0.0000	0.5000
220	0.2857	0.6125
245	1.0000	0.8413
280	2.0000	0.9772

i Solution

part	means	shape	read from the table as	probability
a	X < 175	left tail	P(X <= 175)	0.1587
b	X > 245	right tail	1 - P(X <= 245)	0.1587
c	175 < X < 245	straddles the mean	P(X <= 245) - P(X <= 175)	0.6827
d	220 < X < 280	entirely above the mean	P(X <= 280) - P(X <= 220)	0.3648
e	X > 140	right tail	1 - P(X <= 140)	0.9772
f	X = 210	a single point	zero	0.0000

Part (c) is the empirical rule's 68%, since \$175 and \$245 are one standard deviation either side of the mean. Parts (a) and (b) are equal by symmetry. Part (d) is the one that lies entirely above the mean: subtract two cumulative probabilities, and do not add 0.5 to anything.

Part (f) is zero, because a continuous random variable has no probability at a single point. If the manager means "\$210 to the nearest dollar," that is the interval from \$209.50 to \$210.50, with probability 0.0114. Asking what a question *means* is worth doing before computing anything.

Working backwards: percentiles and targets

The questions so far supply a value and ask for a probability. Managers ask the reverse just as often: *what value* puts a given share of outcomes below it? That is a **percentile**, and finding one means running the cumulative probability backwards.

To find the value x with a share p of the distribution below it:

1. Find the z-score with area p to its left: `=NORM.S.INV(p)`.
2. Convert it back to the original units with the z-score formula solved for x:

$$x = \mu + z\,\sigma.$$

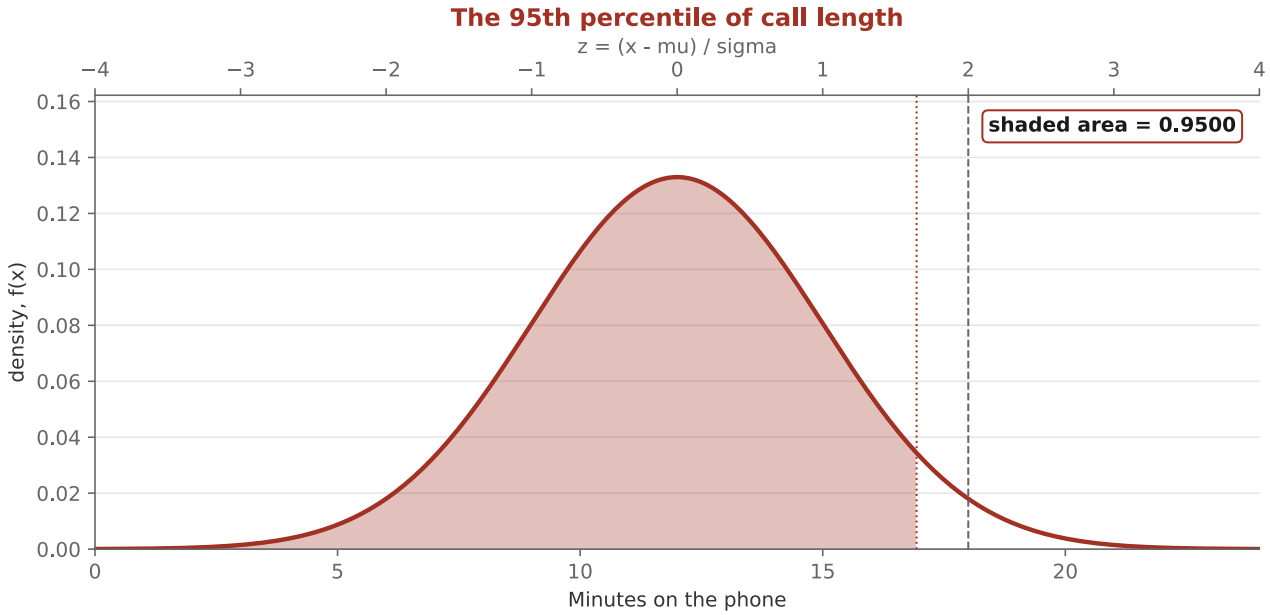
Or in one step: `=NORM.INV(p, mu, sigma)`.

Application: a service-level goal

Management at the cable company wants 95% of calls to be finished in under 18 minutes. With calls normal, mean 12 and standard deviation 3, is the goal being met?

share below	z = NORM.S.INV(share)	x = mu + z sigma	Excel
0.95	1.6449	16.9346	=NORM.INV(0.95, 12, 3)

$$x = \mu + z\sigma = 12 + (1.6449)(3) = 16.93 \ \text{ minutes}$$



Ninety-five percent of calls finish within 16.93 minutes, which is inside the 18-minute target, so the goal is met. There are two equivalent ways to check a goal like this, and it is worth knowing both:

- compare the **95th percentile** (16.93 minutes) with the **target** (18 minutes), or
- compare the **share of calls under the target** (97.72%) with the **required share** (95%).

They always agree. Use whichever matches how the question is asked.

Application: a reorder point

A Henderson gas station waits 4 days for each delivery of regular gasoline. Demand during those 4 days, the **lead time**, is normal with a mean of 930 gallons and a standard deviation of 140. The station reorders when its tank falls to the **reorder point**. Where should the reorder point be for a 90% chance of not running out before the delivery arrives? What about 95% or 99%?

share below	z = NORM.S.INV(share)	x = mu + z sigma	Excel	safety stock (x - mu)
0.90	1.28	1109.42	=NORM.INV(0.9, 930, 140)	179.42
0.95	1.64	1160.28	=NORM.INV(0.95, 930, 140)	230.28
0.99	2.33	1255.69	=NORM.INV(0.99, 930, 140)	325.69

For a 90% service level, reorder at 1,109 gallons. The 179 gallons above average lead-time demand are the **safety stock**, and in this model the safety stock is just $z\sigma$.

> Read down the safety stock column. Going from 90% to 95% costs 51 more gallons. Going from 95% to 99% costs 95 more, for a smaller gain in protection. Each additional point of service level is more expensive than the last, because you are buying your way further into a tail that keeps getting thinner. That is a business conclusion, and it comes straight from the shape of the curve.
>
> The same table works in reverse. The station's current reorder point is 1,200 gallons. The probability that lead-time demand stays below it is 97.31%: that is its current service level.

The empirical rule, computed instead of remembered

In Chapter 3 the empirical rule was a rule of thumb: about 68%, 95%, and 99.7% of a bell-shaped distribution lie within one, two, and three standard deviations of the mean. Now we can compute it.

within	normal: inside	normal: each tail	Chebyshev: at least inside
mu +/- 1 sigma	0.6827	0.1587	0.0000
mu +/- 2 sigma	0.9545	0.0228	0.7500
mu +/- 3 sigma	0.9973	0.0013	0.8889

For the call centre, that means about 68% of calls last between 9 and 15 minutes, about 95% between 6 and 18 minutes, and practically all between 3 and 21 minutes. Each is $\mu \pm k\sigma$.

> The empirical rule is the normal distribution's rule. Chebyshev's Theorem, in the last column, is the rule for *every* distribution, which is why it is so much weaker.
>
> One more thing about Chebyshev, because it is easy to misuse. It bounds the two tails *together*: at most $1/k^2$ of any distribution lies more than k standard deviations from the mean, on one side or the other. Splitting that in half to get a bound on one tail assumes the two tails are equal, which a skewed distribution does not guarantee. Nothing stops all of the outside probability from sitting in one tail. A bound on a single tail that holds for every distribution exists, and it is called **Cantelli's inequality**: at most $1/(1+k^2)$ lies more than k standard deviations *above* the mean.

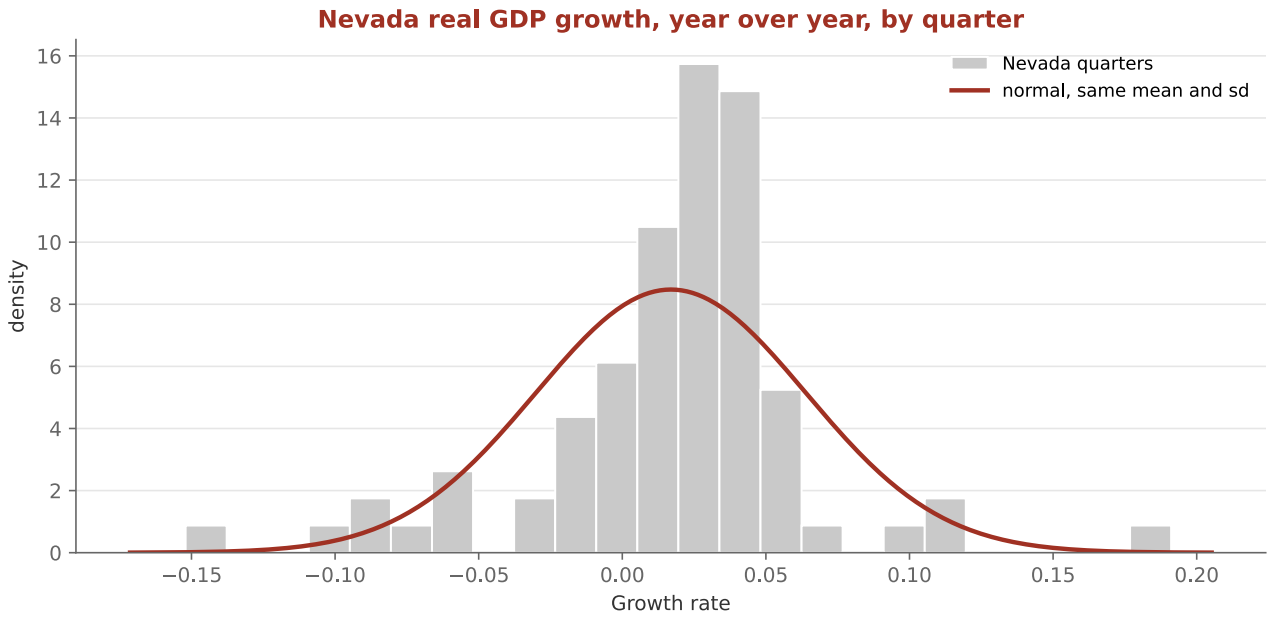
	2 standard deviations above the mean
k (standard deviations above the mean)	2.0000
normal: exact upper tail	0.0228
any distribution (Cantelli): at most	0.2000
any distribution (Chebyshev, both tails): at most	0.2500
half of Chebyshev (NOT a valid bound)	0.1250

For $k = 2$, the normal curve puts 2.28% above the mean plus two standard deviations. A distribution of *unknown* shape can put as much as 20% there. That gap is the price of not knowing the shape, and it is why the next section matters.

Is it normal enough? Nevada's growth rate

Whether data are normal is a claim to be checked, not assumed. In Chapters 4 and 5 we worked with the year-over-year growth rate of Nevada's real GDP, quarter by quarter. Is it normal?

```
80 quarters,  mean = 1.69%,  sd = 4.71%
```



within	interval_low	interval_high	actual_percent	bell_shaped_percent	chebyshev_at_least
1 sd	−0.03	0.06	83.75	68.3	0.00
2 sd	−0.08	0.11	93.75	95.4	75.00
3 sd	−0.12	0.16	97.50	99.7	88.89

below	z	share observed	normal model says	observed / model	observations below
−0.08	−2.0585	0.0500	0.0198	2.5289	4

below	z	share observed	normal model says	observed / model	observations below
−0.05	−1.4212	0.1000	0.0776	1.2882	8
0.00	−0.3591	0.2375	0.3598	0.6602	19
0.05	0.7030	0.9125	0.7590	1.2023	73

The normal curve gets the centre and the tails wrong, in opposite directions. 83.8% of quarters lie within one standard deviation of the mean, where a normal distribution would put 68%. Too many ordinary quarters. At the same time, 2.5% lie beyond three standard deviations, where the normal would put 0.3%. Too many extreme ones.

Read the `observed / model` column of the last table as a risk manager would. The normal model says a quarter of contraction worse than -8% has probability 2.0%. Nevada had 4 such quarters in 80, or 5.0%, about 2.5 times as many. Meanwhile the model puts 36% of quarters below zero growth, and only 24% were.

So a normal model fitted to these data is too *pessimistic* about mild slowdowns and too *optimistic* about severe ones. The second error is the expensive one. A reserve, a credit line, or a staffing plan sized to the normal curve would be sized for a world with fewer deep recessions than Nevada has had.

And look at *when* the eight worst quarters occurred: 2008, 2009, and 2020. Two episodes, the Great Recession and the pandemic, account for all of them. Bad quarters arrive in clumps, the same dependence that broke the binomial in Chapter 5, and part of it is built into a year-over-year measure, where neighbouring quarters share three-quarters of their data. Chapter 12 introduces a formal test of whether data are normal. The habit to build now is to put the curve on top of the histogram and look before you rely on it.

Expected value with normal probabilities

Chapter 5 priced the RAD Construction bridge contract with three subjective probabilities for finishing early, on time, or late. A continuous model gives those probabilities a source.

RAD has been awarded a contract to replace a bridge. From experience on previous projects, RAD believes completion time is normal with a mean of 92 days and a standard deviation of 9 days. It expects a net profit of $125,000. Finishing in less than 84 days earns a $20,000 bonus from the state. Taking longer than 106 days costs a $15,000 penalty. What is RAD's expected profit?

	probability
early (< 84 days)	0.1870
on time (84 to 106 days)	0.7531
late (> 106 days)	0.0599

alternative	payoff if early (< 84 days)	payoff if on time (84 to 106 days)	payoff if late (> 106 days)	EMV	SD	CV
bridge contract	$145,000	$125,000	$110,000	$127,842	$8,956	0.0701

The expected profit is $127,842. The normal model did the part the project manager did by judgment in Chapter 5: it turned a belief about completion time into the three probabilities, a left tail, a right tail, and the interval

between them. The EMV arithmetic is unchanged. If RAD can shave the mean to 88 days, the same two lines of code reprice the contract, which is the practical advantage of a model over a guess.

Focus on analytics: Six Sigma

In 1986 an engineer at Motorola introduced **Six Sigma**, a quality programme whose name comes from the normal distribution: design a process so tightly that the specification limits sit six standard deviations from the target. Six Sigma is famous for promising 3.4 defects per million opportunities. Where does that number come from?

sigma level	DPMO, centred	DPMO, mean shifted 1.5 sigma	yield, shifted
3	2,699.7961	66,810.6	0.9331894
4	63.3425	6,209.7	0.9937903
5	0.5733	232.6	0.9997674
6	0.0020	3.4	0.9999966

> If the process were perfectly centred, being six standard deviations from both limits would produce about 2 defects per *billion*, not 3.4 per million. The famous figure assumes the process mean drifts by up to 1.5 standard deviations over time, which leaves only 4.5 standard deviations to the nearer limit. The normal tail beyond 4.5 standard deviations is 3.4 per million.
>
> Your textbook quotes the 3.4 without the drift, and the number cannot be derived from "six sigma" as stated. The lesson for a manager is general: a headline statistic usually carries an assumption, and you find it by asking the number to be recomputed.

The Normal Approximation to the Binomial

At the end of Chapter 5, the binomial distribution for 50 customers with $p = 0.4$ looked close to a bell curve. That was not an accident. When the number of trials is large enough, a binomial distribution is well approximated by a normal curve with the same mean and standard deviation.

> A binomial distribution with n trials and probability of success p can be approximated by a normal distribution with
>
> $$\mu = np \qquad \text{and} \qquad \sigma = \sqrt{npq}$$
>
> when $np \geq 5$ and $nq \geq 5$, where $q = 1 - p$. In words: when both the expected number of successes and the expected number of failures are at least 5.

Application: drive-through order errors

Carl's Jr. recently reported a 6% error rate on drive-through orders. Here a "success" is an order filled *incorrectly*; success is a label, not a compliment. What is the probability that exactly 7 of the next 120 drive-through orders are wrong?

	Carl's Jr., n = 120, p = 0.06
np	7.2
nq	112.8
np >= 5 and nq >= 5	True

	Carl's Jr., n = 120, p = 0.06
mu = np	7.2
sigma = sqrt(npq)	2.601538

Both conditions hold. The exact answer comes from Chapter 5:

```
BINOM.DIST(7, 120, 0.06, FALSE) = 0.1531
```

To approximate it with the normal curve we run straight into the opening idea of this chapter: under a continuous curve, $P(X = 7) = 0$. A single point has no area. So we give each whole number a slice of the curve one unit wide, centred on it. The number 7 gets the slice from 6.5 to 7.5.



	exactly 7
question	exactly 7
whole numbers	7
normal interval	(6.5, 7.5)
exact binomial	0.153085
normal, corrected	0.151966
normal, uncorrected	0.0
error, corrected	−0.001119
error, uncorrected	−0.153085

The scarlet bar is the exact binomial probability. Its area, one unit wide, and the shaded slice of the curve beneath it, are nearly equal: the approximation gives 0.1520 against an exact 0.1531. Without the half-unit slice, the approximation is zero, which is the last row's point.

The **continuity correction factor** represents each whole number k by the interval from $k - 0.5$ to $k + 0.5$. To approximate a binomial probability:

1. List the whole numbers the question includes.

2. Move the **left** boundary down by 0.5 and the **right** boundary up by 0.5, so each included number gets its whole slice.
3. A boundary at $-\infty$ or $+\infty$ needs no correction.

Translating the question, again

The continuity correction is Chapter 5's phrase guide with one more step. A question about "fewer than 3" and one about "3 or fewer" include different whole numbers, so they get different corrections:

the question says	whole numbers included	normal interval, corrected
fewer than 3	0, 1, ..., 2	X < 2.5
at most 3	0, 1, ..., 3	X < 3.5
exactly 3	3	2.5 < X < 3.5
at least 3	3, 4, ..., n	X > 2.5
more than 3	4, 5, ..., n	X > 3.5
3, 4, or 5	3, ..., 5 (inclusive)	2.5 < X < 5.5

Do the translation in two steps, never one. First, the words become a list of whole numbers: "fewer than 3" is 0, 1, 2. Then the list becomes an interval with half a unit added at each end: $X < 2.5$. Students who go straight from "fewer than 3" to a boundary tend to land on 3.5, because "3" is the number in the sentence. The list of whole numbers is the check.

Here are the textbook's questions about Carl's Jr., answered three ways: exactly, with the correction, and without it.

question	whole numbers	normal interval	exact binomial	normal, corrected	normal, uncorrected	error, corrected	error, uncorrected
exactly 7	7	(6.5, 7.5)	0.1531	0.1520	0.0000	−0.0011	−0.1531
at most 4	0, 1, ..., 4	(-inf, 4.5)	0.1473	0.1497	0.1093	0.0023	−0.0380
more than 8	9, 10, ..., n	(8.5, +inf)	0.2939	0.3086	0.3792	0.0147	0.0853
7 to 9 inclusive	7, ..., 9 (inclusive)	(6.5, 9.5)	0.4007	0.4177	0.2861	0.0170	−0.1146

With the correction, no answer is off by more than 0.017. Without it, the worst is off by 0.153. The half unit is not a rounding nicety.

If software computes the binomial exactly, why learn the approximation? The textbook's reason, saving arithmetic, has expired, as it did for the Poisson approximation in Chapter 5.

The idea has not. A sample proportion is a binomial count divided by n. That it is approximately normal, once n is large enough, is the reason the confidence intervals for proportions in Chapter 8 and the tests in Chapter 9 use z. Every polling margin of error you have ever read rests on this section.

The Harvard Business School website reported that 12% of applicants were admitted in 2016. Suppose 60 applications are selected at random. Use the normal approximation to find the probability that:

(a) fewer than 5 are admitted

(b) 6, 7, 8, or 9 are admitted

(c) 11 or more are admitted

Check that the approximation is licensed, and compare each answer with the exact binomial.

i Solution

```
np = 7.2,  nq = 52.8,  conditions met: True
```

	question	whole numbers	normal interval	exact binomial	normal, corrected	normal, uncorrected	error, corrected	error, uncorrected
a	fewer than 5	0, 1, ..., 4	(-inf, 4.5)	0.1388	0.1417	0.1911	0.0029	0.0522
b	6 to 9 inclusive	6, ..., 9 (inclusive)	(5.5, 9.5)	0.5634	0.5699	0.4459	0.0064	−0.1175
c	at least 11	11, 12, ..., n	(10.5, +inf)	0.0994	0.0949	0.0656	−0.0045	−0.0338

$np = 7.2$ and $nq = 52.8$, so both conditions hold.

(a) Fewer than 5 is 0 through 4, so the boundary is 4.5, not 5.5.

(b) 6 through 9 runs from 5.5 to 9.5.

(c) 11 or more starts at 10.5.

Every corrected answer is within 0.6 percentage points of the exact binomial. The uncorrected answers miss by as much as 12 points, and part (a) shows why the boundary matters: using 5 instead of 4.5 quietly adds most of the slice for “exactly 5” to an event that excludes it.

The Exponential Distribution

The **exponential probability distribution** describes the time (or distance, or other measure) between events: the time between customer arrivals, between machine failures, between website hits. Its density is

$$f(x) = \lambda e^{-\lambda x} \qquad \text{for} \quad x \geq 0,$$

where $e \approx 2.71828$ and λ is the **rate** at which events occur: the average number of events per unit of time. λ can be any positive number. It does not have to be a whole number: 0.25 customers per minute is a perfectly good rate.

The **mean** and the **standard deviation** of an exponential distribution are equal:

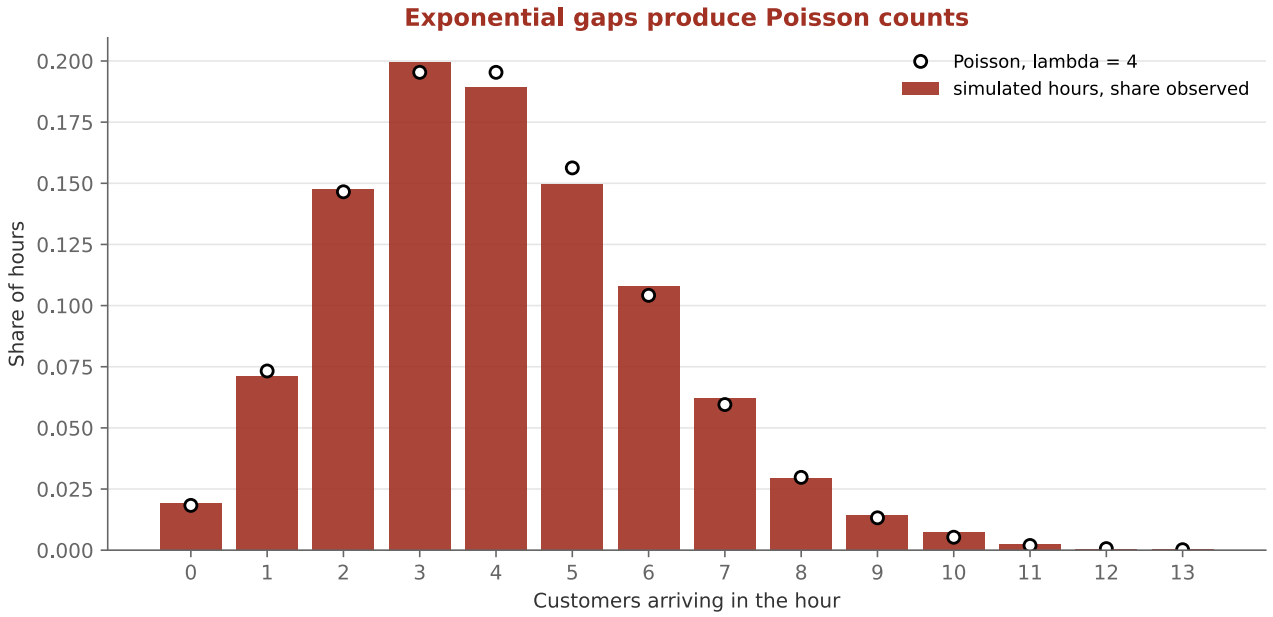
$$\mu = \sigma = \frac{1}{\lambda}.$$

The other side of the Poisson

You have met λ before. In Chapter 5 the Poisson distribution counted the number of arrivals in an interval, with mean λ. The exponential distribution measures the time **between** those same arrivals. It is one process seen two ways: count the events, and you get a Poisson; time the gaps, and you get an exponential.

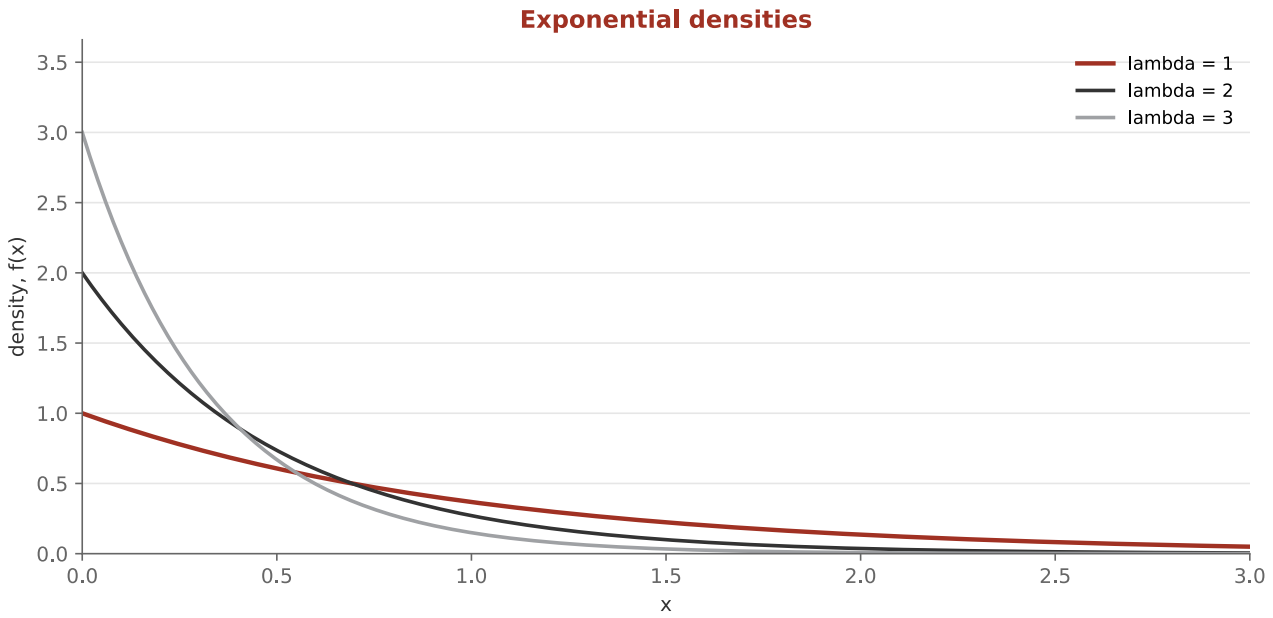
We can watch that happen. Below are 20,000 simulated waiting times between customers at a store that averages 4 customers an hour, drawn from an exponential distribution with $\lambda = 4$ per hour. Lay them end to end, and count how many customers arrive in each hour:

```
20,000 waiting times cover 4,972 complete hours.
Mean waiting time  = 0.2486 hours  (1/lambda = 0.2500)
Arrivals per hour: mean = 4.022, variance = 4.113  (Poisson: both equal lambda = 4)
```



The counts land on the Poisson probabilities from Chapter 5, and their mean and variance are both close to 4. Nothing about counts was put into the simulation. Only the gaps were.

The shape of the exponential



Every exponential curve starts at a height of λ at $x = 0$ and decays toward zero. A higher rate means events come faster, so short waits are more concentrated near zero. Note again that two of these curves start above 1. They are densities.

Three differences between the exponential and the normal:

1. The exponential is **right-skewed**; the normal is symmetric.
2. The exponential is described by **one** parameter, λ (or equivalently $\mu = 1/\lambda$); the normal needs two, μ and σ.
3. An exponential random variable **cannot be negative**. The next customer cannot arrive two minutes *before* the last one. A normal random variable can take any value.

Getting λ and μ straight

The textbook's author reports that choosing the value of λ is where his students struggle most, and it is where most wrong answers in this section start.

- λ is a **countable rate**: events *per* unit of time, such as customers per minute.
- $\mu = 1/\lambda$ is a **measurable interval**: time *per* event, such as minutes per customer.
- Both must be in the **same time unit** as the question. If the question asks about minutes, λ must be per minute.

Three ways the information arrives, and what to do with each:

	lambda (events per minute)	mu = 1/lambda (minutes per event)	sigma = mu (minutes)
deli counter: a customer every 4 minutes	0.2500	4.0000	4.0000
same deli, quoted per hour: 15 customers an hour, question in minutes	0.2500	4.0000	4.0000
hockey: 32.2 shots per 60-minute game, question in minutes	0.5367	1.8634	1.8634

The first two rows are the same deli described two ways, and they give the same $\lambda = 0.25$ per minute. A customer every 4 minutes *is* 15 customers an hour. Whenever the rate is quoted for a different interval than the one the question asks about, the first line of your work is a new λ, as it was for the Poisson in Chapter 5. And when you are handed a mean time between events, the first line is $\lambda = 1/\mu$. Plugging a mean into a formula that wants a rate is the most common error in this section, and it produces an answer that looks perfectly reasonable.

Calculating exponential probabilities

The **exponential cumulative distribution function** gives the probability that the waiting time is at most a:

$$P(x \leq a) = 1 - e^{-\lambda a}.$$

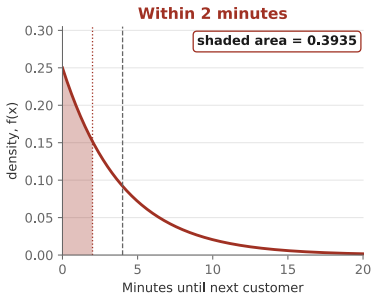
Its complement, the probability of waiting **longer** than a, is simply

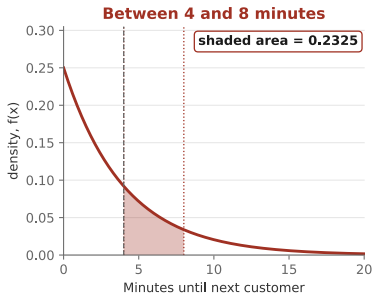
$$P(x > a) = e^{-\lambda a}.$$

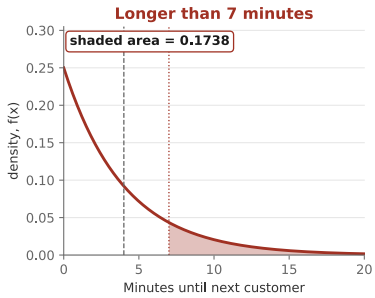
In Excel, `=EXPON.DIST(a, lambda, TRUE)` returns $P(x \le a)$. Note that Excel wants the **rate**, λ, not the mean.

At the deli counter, customers arrive on average every 4 minutes, so $\lambda = 0.25$ per minute. What is the probability that the next customer arrives within 2 minutes? Between 4 and 8 minutes from now? Not for more than 7 minutes?

question	event	P(X <= upper)	P(X <= lower)	computed as	probability
within 2 minutes	X < 2	0.3935	0.0000	1 - e^(-0.25 x 2)	0.3935
between 4 and 8 minutes	4 < X < 8	0.8647	0.6321	P(X <= 8) - P(X <= 4)	0.2325
longer than 7 minutes	X > 7	1.0000	0.8262	e^(-0.25 x 7)	0.1738








```
within 2 minutes          =EXPON.DIST(2, 0.25, TRUE)
between 4 and 8 minutes   =EXPON.DIST(8, 0.25, TRUE) - EXPON.DIST(4, 0.25, TRUE)
longer than 7 minutes     =1 - EXPON.DIST(7, 0.25, TRUE)
```

The dashed line in each chart is the mean, 4 minutes. There is a 39.3% chance of a customer within 2 minutes, a 23.3% chance of one between 4 and 8 minutes, and a 17.4% chance of a wait longer than 7.

The mean is not the typical wait

The average wait at the deli is 4 minutes. The median wait, the time with half of waits shorter, is only 2.77 minutes, and 63.2% of waits are shorter than the mean. In a right-skewed distribution a few long waits pull the mean above the typical case, which is the Chapter 3 lesson about skew applied to a model.

The exponential percentile has a closed form. The wait with a share p of waits shorter than it is

$$x = \frac{-\ln(1-p)}{\lambda}.$$

At $p = 0.5$ this is the median, $\ln 2/\lambda \approx 0.693\,\mu$.

A staffing question in this form: how long a gap between customers should the deli plan for, if it wants to cover 95% of gaps? $-\ln(0.05)/0.25 = 11.98$ minutes, almost three times the mean.

The exponential has no memory. If a customer has not arrived in the last 5 minutes, the probability of waiting more than another 4 minutes is $P(x > 9 \mid x > 5) = e^{-0.25(9)}/e^{-0.25(5)} = e^{-0.25(4)} = 0.3679$,

exactly the same as the probability of waiting more than 4 minutes starting from scratch. No customer is ever "due."

This is a property worth checking against the business. Arrivals of independent customers really do behave this way. Machine parts that wear out do not: a bearing that has run 10,000 hours is *more* likely to fail in the next hour than a new one. If the thing you are timing ages, the exponential is the wrong model, however convenient its formula.

Exercises

These questions all concern an exponential distribution with a **mean** of 0.25 hours between events.

(a) What is λ, per hour?

(b) What is the probability that the next event comes between 0.5 and 0.75 hours from now?

(c) Between 0.15 and 0.35 hours?

(d) After more than 0.15 hours?

(e) Within 0.4 hours?

Solution

```
lambda (events per hour)          4.00
mu = 1/lambda (hours per event)   0.25
sigma = mu (hours)                0.25
```

	event	computed as	probability	Excel
b	0.5 < X < 0.75	P(X <= 0.75) - P(X <= 0.5)	0.0855	=EXPON.DIST(0.75, 4, TRUE) - EXPON.DIST(0.5, 4…
c	0.15 < X < 0.35	P(X <= 0.35) - P(X <= 0.15)	0.3022	=EXPON.DIST(0.35, 4, TRUE) - EXPON.DIST(0.15, ...
d	X > 0.15	e^(-4 x 0.15)	0.5488	=1 - EXPON.DIST(0.15, 4, TRUE)
e	X < 0.4	1 - e^(-4 x 0.4)	0.7981	=EXPON.DIST(0.4, 4, TRUE)

(a) A mean of 0.25 hours between events is $\lambda = 1/0.25 = 4$ events per hour. That is the whole exercise: everything after it is substitution.

(b) Both ends are beyond the mean, in the thin part of the tail, so the probability is small.

(c) This interval straddles the mean of 0.25 hours.

(d) A right tail: $e^{-4(0.15)}$, no subtraction from 1 needed.

(e) Parts (d) and (e) do not sum to 1, because they are not complements: "more than 0.15" and "within 0.4" overlap. Check complements only when the two events really are opposites.

The Uniform Distribution

With the **continuous uniform probability distribution**, every interval of the same width within the range has the same probability. If the smallest possible value is a and the largest is b, the density is

$$f(x) = \begin{cases} \frac{1}{b-a}, a \le x \le b \\ 0, \quad \text{otherwise.} \end{cases}$$

The probability of an interval is its width as a share of the whole range:

$$P(x_1 \le x \le x_2) = \frac{x_2 - x_1}{b - a}.$$

The mean and standard deviation are

$$\mu = \frac{a+b}{2} \qquad \text{and} \qquad \sigma = \frac{b-a}{\sqrt{12}}.$$

Where uniform distributions come from

Uniform distributions are less common in nature than the textbook's examples suggest, but there is one place they arise for a clear reason. A resort shuttle leaves the lobby every 12 minutes, on schedule. A guest who walks out at a random moment, with no idea of the timetable, waits somewhere between 0 and 12 minutes, and no part of that range is more likely than any other. That wait is uniform on $[0, 12]$, because the guest's arrival is unrelated to the schedule.

```
height of the density   f(x) = 1/(b - a) = 0.0833 per minute
mean wait               (a + b)/2        = 6.00 minutes
standard deviation      (b - a)/sqrt(12) = 3.4641 minutes
P(wait more than 9 min) (12 - 9)/12      = 0.2500
```

Compare this with the deli counter. There, customers arrive at random and the wait is exponential, so short waits are most likely. Here the *shuttle* is on a schedule and only the guest is random, so every wait is equally likely. Which distribution describes a waiting time depends on what is random and what is fixed.

Application: a start-up checklist

The time a manager takes to run the start-up checklist for a production line is uniform between 15 and 35 minutes.

	event	computed as	probability
under 20 minutes	15 < X < 20	(20 - 15) / (35 - 15)	0.25
between 20 and 25 minutes	20 < X < 25	(25 - 20) / (35 - 15)	0.25
over 30 minutes	30 < X < 35	(35 - 30) / (35 - 15)	0.25



The density is $1/(35-15) = 0.05$ per minute everywhere in the range. The probability of finishing between 20 and 25 minutes is the area of a rectangle 5 minutes wide and 0.05 high, or 0.25. Any 5-minute stretch inside the range has the same probability.

Working backwards with a uniform

The plant manager wants 80% of start-ups completed within 30 minutes. Is the goal being met?

```
80th percentile  x = a + 0.80 (b - a) = 15 + 0.80 (20) = 31.0 minutes
share finished within 30 minutes      = (30 - 15)/20  = 0.75
```

The 80th percentile is 31 minutes, beyond the 30-minute target, and only 75% of start-ups finish within 30 minutes. The goal is not being met. As with the cable company, the two checks agree.

The mean start-up time is 25 minutes, with a standard deviation of 5.77 minutes. Excel has no uniform distribution function; the formulas are short enough that the arithmetic is the function.

Which Distribution Is It?

As in Chapter 5, nobody tells you which distribution applies. Four questions settle most cases.

Ask	If yes
Is it a measurement that clusters around a centre, symmetric, with extremes rare in both directions? And have you looked at the data?	**Normal**, with μ and σ
Is it the time (or distance) *between* independent events that occur at a steady average rate?	**Exponential**, with λ
Is every value in a fixed range equally likely, typically because something random meets something scheduled?	**Uniform**, with a and b
Is it a *count* of successes in n trials, with $np \geq 5$ and $nq \geq 5$?	**Binomial**, which you may approximate by a **normal** with $\mu = np$, $\sigma = \sqrt{npq}$ and the continuity correction

Name the distribution and its parameters for each of the following.

(a) A help desk receives 18 calls an hour. How long until the next call?

(b) A bottling line fills bottles to a mean of 500 ml with a standard deviation of 2 ml. How much is in the next bottle?

(c) A monorail train leaves the station every 8 minutes. You arrive with no idea of the timetable. How long do you wait?

(d) 3% of a casino's 400 slot machines need service on a given day. How many need service tomorrow?

(e) 40% of 200 surveyed Las Vegas visitors say they will return within a year. How many say so?

(f) How long until a light bulb burns out, when bulbs fail mostly after long use?

i Solution

(a) Exponential. The rate is per hour; if the question is in minutes, $\lambda = 18/60 = 0.3$ per minute and $\mu = 1/0.3 \approx 3.33$ minutes.

(b) Normal, $\mu = 500$, $\sigma = 2$, provided fill weights look bell-shaped when you check them. Fill weights on a well-run line usually do.

(c) Uniform on $[0, 8]$ minutes: a random arrival meeting a fixed schedule.

(d) A binomial count, $n = 400$, $p = 0.03$. Here $np = 12$ and $nq = 388$, so the normal approximation is licensed, with $\mu = 12$ and $\sigma = \sqrt{400(0.03)(0.97)} = 3.41$. The Poisson with $\lambda = 12$ is licensed too, by Chapter 5's rule ($n \geq 20$, $p \leq 0.05$). Two approximations, one count.

(e) A binomial count, $n = 200$, $p = 0.40$, approximately normal with $\mu = 80$ and $\sigma = 6.93$.

(f) It looks like a waiting time, which suggests the exponential, but the exponential has no memory, and a bulb that fails mostly after long use is ageing. The exponential is the wrong model. Statisticians use other distributions (the Weibull is the usual one) that let the failure rate rise with age. Recognizing when the convenient model does not fit is the skill.

Applications

Attempt each one before opening the solution.

Interstate speeds

Speeds on a stretch of I-15 between Las Vegas and Primm are normal with a mean of 72 mph and a standard deviation of 6 mph.

(a) What share of cars are travelling above the 75 mph limit?

(b) What share are travelling between 65 and 80 mph?

(c) The Highway Patrol wants to set an enforcement threshold that only the fastest 5% of cars exceed. What speed is it?

(d) Is the normal model reasonable for speeds on a congested weekend afternoon? Why or why not?

i Solution

```
                event              shape             computed as  probability
question
1               X > 75          right tail           1 - P(X <= 75)       0.3085
2         65 < X < 80  straddles the mean  P(X <= 80) - P(X <= 65)       0.7871
```

share below	z = NORM.S.INV(share)	x = mu + z sigma	Excel
0.95	1.64	81.87	=NORM.INV(0.95, 72, 6)

(a) 30.9% of cars exceed 75 mph, a right tail.

(b) 78.7%. The interval straddles the mean.

(c) The 95th percentile: $72 + 1.645(6) = 81.9$ mph. The question supplies a share and asks for a value, so it runs the cumulative probability backwards.

(d) Probably not. On a congested afternoon, speeds pile up near a ceiling set by traffic, with a long tail of very slow cars in the jams. That is a left-skewed distribution, and the normal model would misstate both tails. The parameters came from somewhere: ask whether they came from conditions like the ones you are asking about.

The ATM

Customers arrive at a casino-floor ATM at an average rate of 15 an hour. Assume the time between arrivals is exponential.

(a) What are λ per minute and the mean time between arrivals?

(b) What is the probability that the next customer arrives within 3 minutes?

(c) In more than 7 minutes?

(d) Between 4 and 8 minutes?

(e) What is the median time between customers?

i Solution

```
lambda (customers per minute)          0.25
mu = 1/lambda (minutes per customer)   4.00
sigma = mu (minutes)                   4.00
```

	event	computed as	probability	Excel
b	X < 3	1 - e^(-0.25 x 3)	0.5276	=EXPON.DIST(3, 0.25, TRUE)
c	X > 7	e^(-0.25 x 7)	0.1738	=1 - EXPON.DIST(7, 0.25, TRUE)
d	4 < X < 8	P(X <= 8) - P(X <= 4)	0.2325	=EXPON.DIST(8, 0.25, TRUE) - EXPON.DIST(4, 0.2…

(a) 15 an hour is $\lambda = 15/60 = 0.25$ per minute, a mean of 4 minutes between customers. This is the deli counter in disguise.

(b) to **(d)** are read from the table. Part (c) is a right tail, $e^{-0.25(7)}$.

(e) $\ln 2/0.25 = 2.77$ minutes, well below the 4-minute mean.

Airport security

Suppose the time to pass through a security checkpoint at Harry Reid International is uniform between 5 and 30 minutes.

(a) What is the height of the density, and what are the mean and standard deviation?

(b) What is the probability that the next passenger needs less than 12 minutes? More than 20 minutes? Exactly 15 minutes?

(c) What time represents the 75th percentile?

(d) Would you believe a uniform model for this? What would you expect the real distribution to look like?

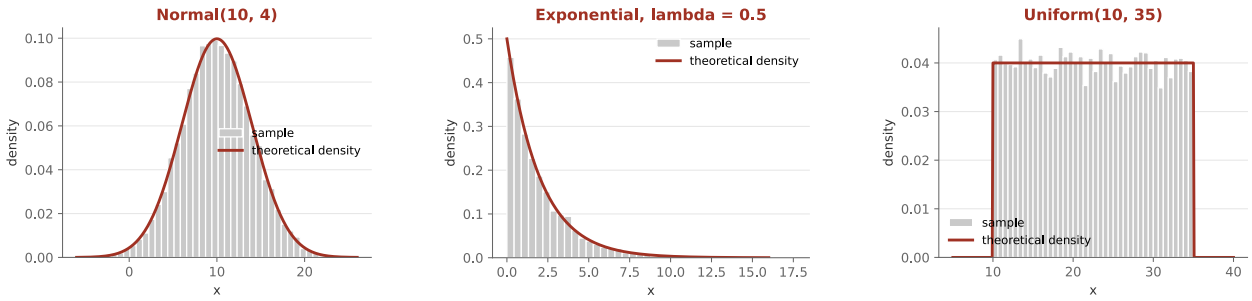
i Solution

```
(a) f(x) = 1/(30 - 5) = 0.04;  mean = 17.5;  sd = 7.2169
(b) P(x < 12) = (12 - 5)/25 = 0.28
    P(x > 20) = (30 - 20)/25 = 0.40
    P(x = 15) = 0, a single point
(c) 75th percentile = 5 + 0.75(25) = 23.75 minutes
```

(d) Hard to believe. A uniform model says a 6-minute wait is exactly as likely as a 29-minute wait, and that nobody ever waits 31. Real queues produce right-skewed waiting times: most people get through fairly quickly, and a minority wait a long time when lines build. The uniform model is the textbook's convenience. When a question hands you a distribution, ask whether you would have chosen it.

Statistical Laboratory

The distributions in this chapter are **data generating mechanisms**: rules that produce samples. Each section above computed what the mechanism implies. Here we run each mechanism and compare the sample with the theory.



	sample mean	true mean	sample sd	true sd
normal	10.0118	10.0	3.9958	4.0000
exponential	1.9830	2.0	1.9688	2.0000
uniform	22.4323	22.5	7.2210	7.2169

Each histogram is drawn on the density scale, so its bars have total area 1 and can be laid under the curve. The samples follow their curves, and the sample means and standard deviations are close to the true values. For the exponential, note what "true" means: with $\lambda = 0.5$, both the mean and the standard deviation are $1/\lambda = 2$, not 0.5.

How close is close, and how does it depend on the sample size? Below, for each sample size, we draw 2,000 samples and record how far the sample mean and the sample standard deviation land from the truth, on average.

	sample size	true mean	average |sample mean - mu|	true sd	average |sample sd - sigma|
normal(10, 4)	10	10.0	1.0001	4.0000	0.7417
	100	10.0	0.3198	4.0000	0.2269
	1000	10.0	0.1033	4.0000	0.0733
	10000	10.0	0.0331	4.0000	0.0223
exponential(0.5)	10	2.0	0.5001	2.0000	0.6162
	100	2.0	0.1652	2.0000	0.2257
	1000	2.0	0.0492	2.0000	0.0718
	10000	2.0	0.0157	2.0000	0.0225
uniform(10, 35)	10	22.5	1.8541	7.2169	0.9238
	100	22.5	0.5737	7.2169	0.2644
	1000	22.5	0.1812	7.2169	0.0803
	10000	22.5	0.0586	7.2169	0.0252

> In every case the average error shrinks as the sample grows, and it shrinks in a regular way: a sample 100 times larger has errors about 10 times smaller. That pattern, error proportional to $1/\sqrt{n}$, is the subject of Chapter 7, and it is the reason a sample of a few thousand voters can describe a country. Notice also that the standard deviation is estimated less precisely than the mean, and least precisely for the skewed exponential, where the occasional very long wait moves the sample standard deviation around.

Appendix: Where the Formulas Come From

You will not be asked to derive these. They are here for anyone who wants to see that the formulas in the blue boxes are consequences of the densities, not additional facts to memorize. For a continuous random variable, the mean is the density-weighted average of x:

$$E[x] = \int x\, f(x)\, dx,$$

the continuous version of $\sum x\, P(x)$ from Chapter 5, and the variance is $E[x^2] - (E[x])^2$, the continuous version of Chapter 5's shortcut formula.

The uniform distribution

With $f(x) = 1/(b-a)$ on $[a, b]$,

$$E[x] = \int_a^b \frac{x}{b-a}\, dx = \frac{b^2 - a^2}{2(b-a)} = \frac{(b-a)(b+a)}{2(b-a)} = \frac{a+b}{2},$$

$$E[x^2] = \int_a^b \frac{x^2}{b-a}\, dx = \frac{b^3 - a^3}{3(b-a)} = \frac{a^2 + ab + b^2}{3},$$

$$\sigma^2 = \frac{a^2 + ab + b^2}{3} - \frac{(a+b)^2}{4} = \frac{4a^2 + 4ab + 4b^2 - 3a^2 - 6ab - 3b^2}{12} = \frac{(b-a)^2}{12}.$$

So $\sigma = (b-a)/\sqrt{12}$.

The exponential distribution

With $f(x) = \lambda e^{-\lambda x}$ on $[0, \infty)$, integrate by parts with $u = x$ and $dv = \lambda e^{-\lambda x} dx$, so that $du = dx$ and $v = -e^{-\lambda x}$:

$$E[x] = \int_0^\infty x\, \lambda e^{-\lambda x}\, dx = \Big[-x e^{-\lambda x}\Big]_0^\infty + \int_0^\infty e^{-\lambda x}\, dx = 0 + \Big[-\frac{1}{\lambda} e^{-\lambda x}\Big]_0^\infty = \frac{1}{\lambda}.$$

The same step applied twice gives $E[x^2] = 2/\lambda^2$, so $\sigma^2 = 2/\lambda^2 - 1/\lambda^2 = 1/\lambda^2$ and $\sigma = 1/\lambda$: the mean and the standard deviation are equal. The cumulative probability is the area from 0 to a:

$$P(x \le a) = \int_0^a \lambda e^{-\lambda x}\, dx = \Big[-e^{-\lambda x}\Big]_0^a = 1 - e^{-\lambda a}.$$

Setting $1 - e^{-\lambda x} = p$ and solving gives the percentile, $x = -\ln(1-p)/\lambda$.

The normal distribution

Substitute $z = (x - \mu)/\sigma$, so $x = \mu + \sigma z$ and the density becomes the standard normal density $\phi(z) = e^{-z^2/2}/\sqrt{2\pi}$:

$$E[x] = \int_{-\infty}^{\infty} (\mu + \sigma z)\, \phi(z)\, dz = \mu \int_{-\infty}^{\infty} \phi(z)\, dz + \sigma \int_{-\infty}^{\infty} z\, \phi(z)\, dz = \mu \cdot 1 + \sigma \cdot 0 = \mu.$$

The first integral is the total area under a density, which is 1. The second is 0 by symmetry: $z\,\phi(z)$ takes equal and opposite values at z and $-z$, so the area to the left of zero cancels the area to the right. No special functions are needed.

Laboratory 6

There are two ways to get these files to Claude. **Start with Path 1.** It works on every Claude account, including the free tier, and needs nothing turned on.

Path 1 — upload the files

Download the Laboratory 6 pack — one file. Unzip it and you will have a folder holding 7 files. Drag all 7 into a new Claude conversation, wait for each to show as attached, then paste this:

> **PROMPT**
>
> I have uploaded 7 files for a graduate business statistics course. Read `COURSE_CONTEXT.md` first — it explains how this course expects analytical work to be done.
>
> Then run `06_continuous_distributions.py` and show me its actual output, verbatim. Walk me through what each section does and what the results mean. Do not report any number you did not compute, and if anything fails, show me the error rather than describing what the output would have been.

Path 2 — let Claude fetch the files

If web access is working on your account, Claude can retrieve the files itself:

> **PROMPT**
>
> Open https://crooker.faculty.unlv.edu/mba775/fall2026/upload_packs/ — it is a plain page listing the files for each laboratory. Follow the links for Laboratory 6 and read `COURSE_CONTEXT.md` first; it explains how this course expects analytical work to be done.
>
> Then run `06_continuous_distributions.py` and show me its actual output, verbatim. Walk me through what each section does and what the results mean. Do not report any number you did not compute, and if anything fails, show me the error rather than describing what the output would have been.
>
> If you cannot open that page, tell me plainly and I will upload the files instead.

If Claude answers that it cannot reach the page, **nothing is wrong with your work or with the repository.** Claude's web access is a setting, and even when it is on, Claude cannot always retrieve a file by direct link. Use Path 1 instead — it is the ordinary route, not a consolation prize.

Your own junior analyst

The lab script answers the questions in this note. The skill to practice is getting a *new* question answered and checking the answer. Try this, with your own numbers in place of mine:

> **PROMPT**
>
> A casino cage averages 45 customers per hour, and the time between customers is exponential. By running code, find (1) the probability that the next customer arrives within 30 seconds, and (2) the time between customers that 90% of gaps are shorter than. Before you compute anything, tell me the value of lambda you are using, in which time unit, and why. Then show the code, its output, and the equivalent Excel formula for (1). Draw the density with the area for (1) shaded, and label the vertical axis correctly.

Check four things in the reply: that λ was converted to the time unit of the question (30 seconds is half a minute, or 1/120 of an hour), that the Excel formula passes the **rate** to `EXPON.DIST` and not the mean, that the vertical axis of the chart says density rather than probability, and that the numbers came from executed code.

A second one, for the normal distribution:

PROMPT

Daily covers (meals served) at a Summerlin restaurant are normal with a mean of 340 and a standard deviation of 45. By running code, find the probability of more than 400 covers, of between 300 and 360, and the number of covers the restaurant should prep for to meet demand on 95% of days. For each one, say which shape of interval it is (left tail, right tail, straddles the mean, entirely above or below the mean), give the Excel formula, and shade it on a chart.

Working in Excel

PROMPT

Build me an Excel workbook with three sheets, one each for the normal, exponential, and uniform distributions. On each sheet, the parameters (mean and standard deviation; lambda; or a and b) are editable input cells at the top, along with a lower and an upper bound for a question. Below them, compute P(X < upper), P(X > lower), and P(lower < X < upper) with real formulas that reference the input cells (NORM.DIST and EXPON.DIST, and the width formula for the uniform), plus the mean, the standard deviation, and a cell where I type a share p and get the percentile back (NORM.INV, and the closed-form formulas for the other two). Add a line chart of the density over a sensible range, with the vertical axis titled “density”. Pre-fill the normal sheet with mean 12 and standard deviation 3 from my Chapter 6 lecture notes. Give me the file to download.

On the normal sheet, set the bounds to 9 and 15, then 6 and 18, then 3 and 21, and confirm the empirical rule’s 68%, 95%, and 99.7%. Then set the standard deviation to 6 and watch the chart flatten while the area stays 1. Answer in one paragraph: if a manager told you she wanted the curve to be “higher” so that more calls fell near 12 minutes, which parameter would she need to change, and in which direction?